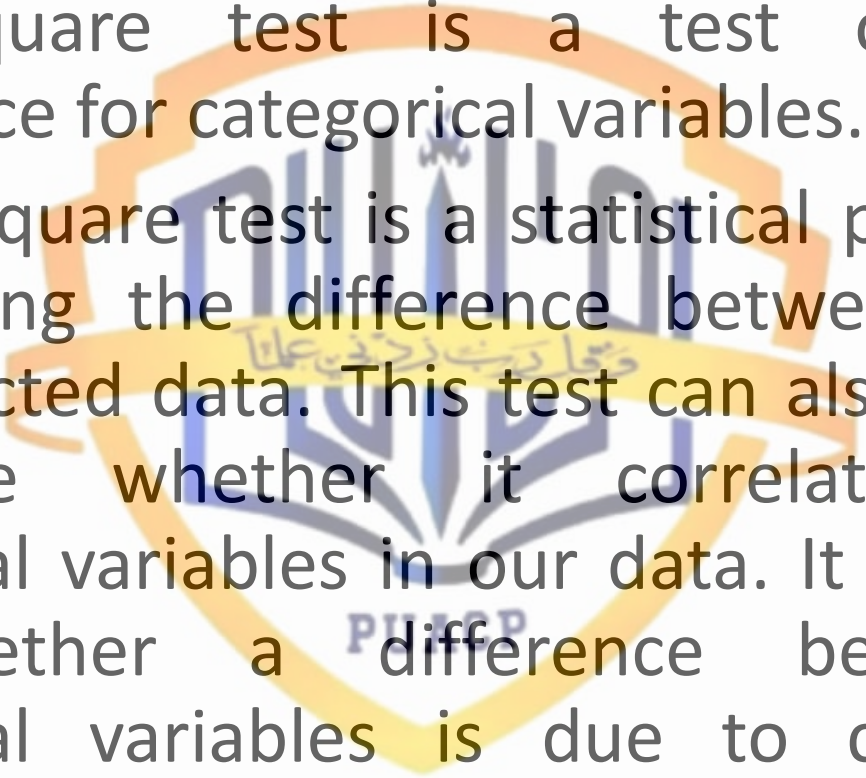


Chi-Square test for Categorical Data

Lecture 7

PUACP

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- The logo of the Punjab University of Journalism and Public Relations (PUJPP) is centered in the background. It features a shield-like emblem with a blue and yellow color scheme. Inside the shield, there is a stylized open book with Arabic calligraphy above it. The acronym 'PUJPP' is written in blue capital letters at the bottom of the shield. The entire logo is overlaid on a faint, light-colored grid.
- A Chi-Square test is a test of statistical significance for categorical variables.
 - The Chi-Square test is a statistical procedure for determining the difference between observed and expected data. This test can also be used to determine whether it correlates to the categorical variables in our data. It helps to find out whether a difference between two categorical variables is due to chance or a relationship between them.

Assumptions

1. The χ^2 assumes that the data for the study is obtained through random selection, i.e. they are randomly picked from the population.
2. The categories are mutually exclusive i.e. each subject fits in only one category. For e.g.- If a person is student of BS-Statistics, he can't be enrolled in BS-Botany.
3. The data should be in the form of frequencies or counts of a particular category and not in percentages.
4. The data should not consist of paired samples or groups or we can say the observations should be independent of each other.

Uses of Chi-Square Test

1. Testing variance of single sample.
2. The Chi-squared test can be used to see if your data follows a well-known theoretical probability distribution like the Normal or Poisson distribution.(Goodness of Fit)
3. Test the association between different categorical variables.

Limitation of Chi-Square Test

1. Chi-squared tests of independence merely indicate the degree of evidence of association.
2. They are rarely adequate for answering all questions about a data set.
3. The chi-squared tests also have limitations in the types of data to which they apply.
4. For instance, they require large samples.
5. This implies that they treat both classifications as nominal.
6. When at least one variable is ordinal, test statistics that utilize the ordinality are usually more appropriate.

Example

Let's say you want to know if **gender** has anything to do with **political party preference**. You poll 440 **voters** in a simple random sample to find out **which political party they prefer**. The results of the survey are shown in the table below:



	Republican	Democrat	Independent	Total
Male	100	70	30	200
Female	140	60	20	220
Total	240	130	50	440

To see if gender is linked to **political** party preference, perform a Chi-Square test of independence using the steps below.

1. Hypothesis:

H0: There is no link between gender and political party preference.

H0: There is no association between gender and political party preference.

H0: The gender and political party preference are independent of each other.

2. Level of significance:

$$\alpha=0.05$$

3. Test Statistics:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

χ^2 = chi squared

O_i = observed value

E_i = expected value

If it is 2x2 table calculations will be simplified by using

$$\chi^2 = \frac{(ad - bc)^2 / n}{(a + b)(c + d)(a + c)(b + d)}$$

4. Calculations:

$$Ei = \frac{(\text{row Total})(\text{column Total})}{\text{Total number of observations}}$$

The expected value of Male republicans is

$$Ei = \frac{(240)(200)}{440} = 109$$

Expected Values				
	Republican	Democrat	Independent	Total
Male	109	59	22.72	200
Female	120	65	25	220
Total	240	130	50	440

	$(O - E)^2 / E$			
	Republican	Democrat	Independent	Total
Male	0.74311927	2.050847	2.332676056	200
Female	3.333333333	0.384615	1	220
Total	240	130	50	440

χ^2 is the sum of all the values
 $= 0.743 + 2.05 + 2.33 + 3.33 + 0.384 + 1$
 $= 9.837$

5. Critical Value

The critical value required Degree of freedom. The degrees of freedom in this case are equal to the table's number of columns minus one multiplied by the table's number of rows minus one, or $(r-1)(c-1)$. We have $(3-1)(2-1) = 2$.

6. Conclusion

By comparing our obtained statistic to the critical statistic found in the chi-square table. For an alpha level of 0.05 and two degrees of freedom, the critical statistic is 5.991, which is less than our obtained statistic of 9.83. The null hypothesis is falling in critical region, because the critical statistic is higher than your obtained statistic.

Critical values of the Chi-square distribution with d degrees of freedom

Probability of exceeding the critical value

d	0.05	0.01	0.001	d	0.05	0.01	0.001
1	3.841	6.635	10.828	11	19.675	24.725	31.264
2	5.991	9.210	13.816	12	21.026	26.217	32.910
3	7.815	11.345	16.266	13	22.362	27.688	34.528
4	9.488	13.277	18.467	14	23.685	29.141	36.123
5	11.070	15.086	20.515	15	24.996	30.578	37.697
6	12.592	16.812	22.458	16	26.296	32.000	39.252
7	14.067	18.475	24.322	17	27.587	33.409	40.790
8	15.507	20.090	26.125	18	28.869	34.805	42.312
9	16.919	21.666	27.877	19	30.144	36.191	43.820
10	18.307	23.209	29.588	20	31.410	37.566	45.315

Practice Question

Example of the Chi-square test

Suppose you wish to classify defects in the furniture produced by a manufacturing plant based on the type of defects and the production shift. A total of 390 furniture defects were recorded, and the defects were classified as one of four types A, B, C, and D. At the same time, each piece of defected furniture was identified according to the production shift.

Shift	Type of defect			
	A	B	C	D
1	15	21	45	13
2	26	31	34	5
3	33	17	49	20

- Check whether the type of defect and production shift is associated?